Plant nutrition

Photosynthesis: The process by which plants synthesise carbohydrates from raw materials using energy from light.

Word equation:

$$\text{Carbon dioxide} + \text{Water} \xrightarrow[\text{Chlorophyll}]{\text{light}} \text{Glucose} + \text{Oxygen}$$

Chemical equation:

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6O_2$$

Chlorophyll: The green pigment in chloroplasts that traps light energy from sun and converts it to chemical energy to synthesise carbohydrates and store them.

Uses of carbohydrates made by photosythesis:

* Stored as startch because startch is insoluble and unreactive
* To make cellulose cell walls
* Glucose is used in respiration to provide energy
* Sucrose for transport in plant
* Nectar to attract insects for pollination

→ Nitrates or ammonium provide plants with nitrogen which is needed to make proteins for growth and repair. It is responsible for making amino acids.

→ Magnesium ions are needed to make chlorophyll which is needed for photosynthesis which produces glucose.

* Minerals are absorbed by active transport; which is the movement of particles from low concentration to high concentration using energy from respiration with the aid of protein carriers.

Testing leaves for startch

1- Take a leaf from a healthy plant

2- Boil the leaf in water to destroy the cell wall

3- Soak the leaf in ethanol to remove Chlorophyll as chlorophyll is insoluble in water but soluble in ethanol

4- Spread the leaf on a white tile for a clear color change to show

5- Add iodine solution

6- If the leaf has startch, iodine solution turns from reddish brown to blue-black

Is chlorophyll necessary for photosynthesis.

*A variegated leaf (a leaf with only patches of chlorophyll) is used.

* The green part of the plant is the control

* The white part of the plant is the experiment.

After destarching the leaf while on the plant, the plant is exposed to daylight for a few hours. Then, the plant is tested for startch.

Result: Only parts with chlorophyll on them will turn the red-brown iodine solution to blue-black. The white parts stay brown.

Is light necessary for photosynthesis?

- Destartch the plant by putting it in a dark place for a period of time.

- Cover part of the leaf with a black paper. Put it in a warm sunny place and provide enough water & CO_2

- Test the leaf for startch by adding iodine solution.

- The uncovered part of the plant turns from reddish-brown to blue-black

- The covered part of the plant remains brown

Is carbon dioxide necessary for photosynthesis?

- Water two destarched plants growing in pots and cover their shoots in polythene bags.

- Place a dish of soda lime in one pot to absorb the carbon dioxide from the air

- Place a dish of sodium hydrogen carbonate solution in the other pot to produce carbon dioxide (the control)

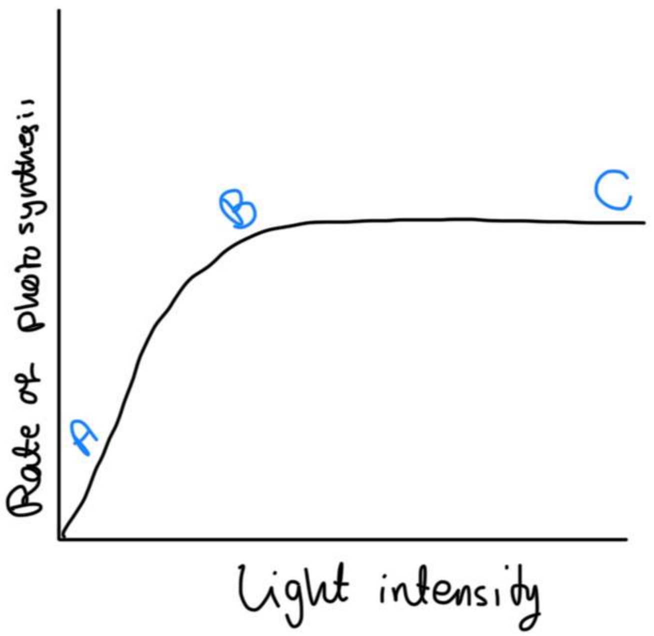
- Place both plants in light for several hours and test a leaf from each plant for starch

→ The leaf that had no carbon dioxide will remain brown
→ The leaf that had carbon dioxide will turn the iodine solution from reddish-brown to blue-black.

Limiting factor : An environmental variable present in a restricted amount in the environment and thus, restricts the effect of other factors.

* Light intensity
* Carbon dioxide concentration
* Temperature
* Water supply

Light intensity

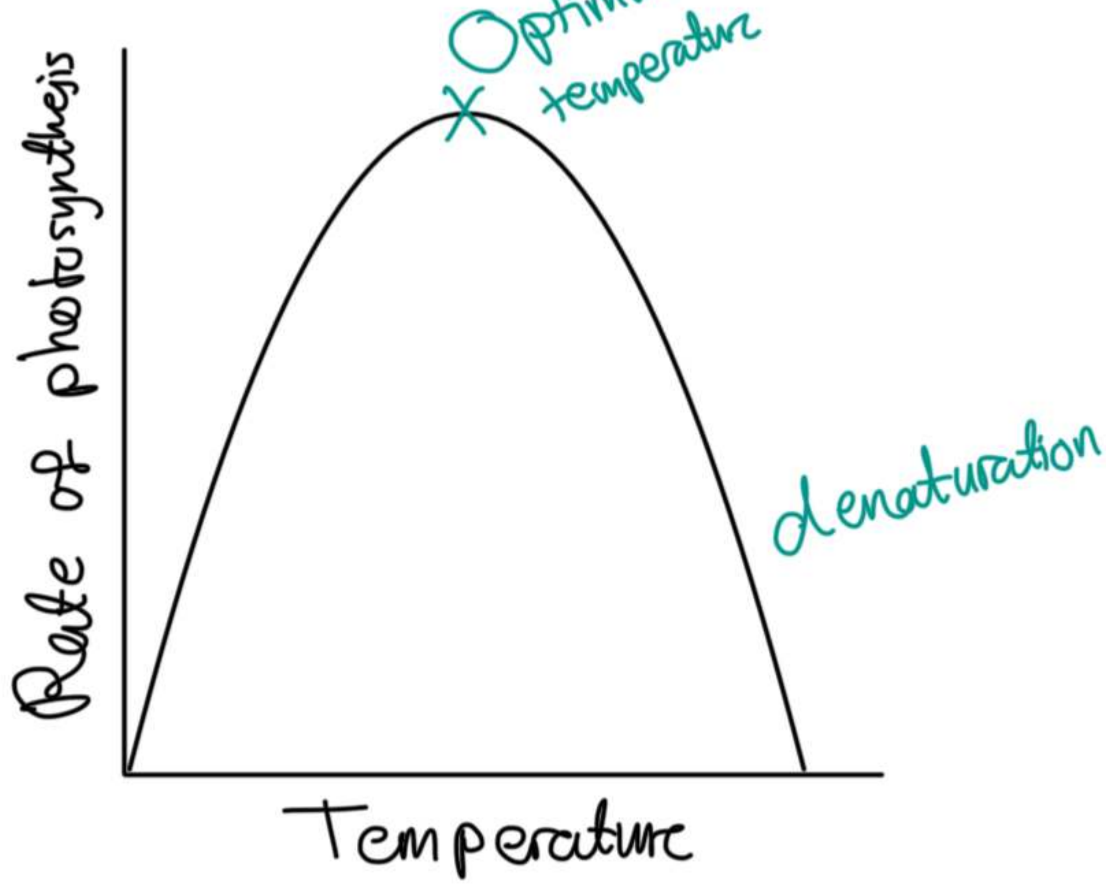


A: As light intensity increases, the rate of photosynthesis increases, up until the plant is doing maximum photosynthesis. Light is the limiting factor at this point, as light would provide energy for photosynthesis. As light intensity increases, more light will be trapped by chlorophyll.

B: Photosynthesis is at its maximum so even if light intensity increases, rate of photosyntesis remains the same. Light intensity stops being the limiting factor at this point

C: Another factor starts being the limiting factor and starts affecting the rate of photosynthesis such as CO_2 concentration or temperature

Temperature



- As temperature increases, photosynthesis rate increase since the enzyme activity increases, up until the optimum temperature for maximum enzyme activity is reached. Temperature is the limiting factor

- As the temperature increases above the optimum temperature, the rate of photosynthesis decreases as enzymes become denatured. Temperature is no longer the limiting factor.

Leaf structure

Cuticle: Made of wax, waterproofing layer. Secreted by cells in the upper epidermis

Upper epidermis: A thin layer of closely fitted cells that do not have any chloroplasts. It acts as a barrier to disease organisms

Palisade mysophyll: The main region for photosynthesis. Cells are tightly packed and contain a lot of chloroplasts to trap light. They recieve carbon dioxide by diffusion from air spaces in the spongy mysophyll

Spongy mysophyll: The cells are spherical and loosely packed. They contain chloroplasts, but not as much as in palisade mysophyll. Air spaces between cells allow gaseous exchange: Carbon dioxide to cells, oxygen from cells.

Vascular bundle: A leaf vein made up of xylem and phloem. Xylem vessels bring water and minerals from roots to leaves. Phloem vessels transport sugars and amino acids from roots $\rightleftharpoons$ leaves

Stomata: Each stoma is surrounded by a pair of guard cells. They swell up to open the stomata, and lose water when closing it. Water vapour passes out during transpiration. Carbon dioxide diffuses in and oxygen diffuses out during photosynthesis

lower epidermis: Acts as a protective layer. Stomata are present to regulate the loss of water. It's the site of gas exchange and out of the leaf